

- 61.** The side BC of a $\triangle ABC$ is produced to D , bisectors of the $\angle ABC$ and $\angle ACD$ meet at P . If $\angle BPC = x^\circ$ and $\angle BAC = y^\circ$, then which one of the following option is correct? **2013 II**
 (a) $x^\circ = y^\circ$ (b) $x^\circ + y^\circ = 90^\circ$
 (c) $x^\circ + y^\circ = 180^\circ$ (d) $2x^\circ = y^\circ$
- 62.** The side AC of a $\triangle ABC$ is produced to D such that $BC = CD$. If $\angle ACB$ is 70° , then what is $\angle ADB$ equal to? **2013 II**
 (a) 35° (b) 45° (c) 70° (d) 110°
- 63.** The heights of two trees are x and y , where $x > y$. The tops of the trees are at a distance z apart. If s is the shortest distance between the trees, then what is s^2 equal to? **2013 II**
 (a) $x^2 + y^2 - z^2 - 2xy$ (b) $x^2 + y^2 - z^2$
 (c) $x^2 + y^2 + z^2 - 2xy$ (d) $z^2 - x^2 - y^2 + 2xy$
- 64.** In a $\triangle ABC$, $\angle B = 90^\circ$ and $\angle C = 2\angle A$, then what is AB^2 equal to? **2013 II**
 (a) $2BC^2$ (b) $3BC^2$ (c) $4BC^2$ (d) $5BC^2$
- 65.** PQR is an equilateral triangle. O is the point of intersection of altitudes PL , QM and RN . If $OP = 8$ cm, then what is the perimeter of the $\triangle PQR$? **2013 II**
 (a) $8\sqrt{3}$ cm (b) $12\sqrt{3}$ cm (c) $16\sqrt{3}$ cm (d) $24\sqrt{3}$ cm
- 66.** $\triangle DEF$ is formed by joining the mid-points of the sides of $\triangle ABC$. Similarly, a $\triangle PQR$ is formed by joining the mid-points of the sides of the $\triangle DEF$. If the sides of the $\triangle PQR$ are of lengths 1, 2 and 3 units, what is the perimeter of the $\triangle ABC$? **2013 II**
 (a) 18 units (b) 24 units
 (c) 48 units (d) Cannot be determined
- 67.** E is the mid-point of the median AD of a $\triangle ABC$. If BE produced meets the side AC at F , then CF is equal to **2013 II**
 (a) $AC/3$ (b) $2AC/3$ (c) $AC/2$ (d) None of these
- 68.** If AD is the internal angular of $\triangle ABC$ with $AB = 3$ cm and $AC = 1$ cm, then what is $BD : BC$ equal to? **2014 I**
 (a) 1 : 3 (b) 1 : 4 (c) 2 : 3 (d) 3 : 4
- 69.** In a $\triangle ABC$, AD is perpendicular to BC and BE is perpendicular to AC . Which of the following is correct? **2014 I**
 (a) $CE \times CB = CA \times CD$ (b) $CE \times CA = CD \times CB$
 (c) $AD \times BD = AE \times BE$ (d) $AB \times AC = AD \times BE$
- 70.** The sides of a triangle are in geometric progression with common ratio $r < 1$. If the triangle is a right angled triangle, the square of common ratio is given by **2014 I**
 (a) $\frac{\sqrt{5} + 1}{2}$ (b) $\frac{\sqrt{5} - 1}{2}$ (c) $\frac{\sqrt{3} + 1}{2}$ (d) $\frac{\sqrt{3} - 1}{2}$
- 71.** If triangles ABC and DEF are similar such that $2AB = DE$ and $BC = 8$ cm, then what is EF equal to? **2014 I**
 (a) 16 cm (b) 12 cm (c) 10 cm (d) 8 cm
- 72.** The sides of a right angled triangle are equal to three consecutive numbers expressed in centimeters. What can be the area of such a triangle? **2014 I**
 (a) 6 cm^2 (b) 8 cm^2 (c) 10 cm^2 (d) 12 cm^2
- 73.** The three sides of a triangle are 15, 25, x units. Which one of the following is correct? **2014 I**
 (a) $10 < x < 40$ (b) $10 \leq x \leq 40$
 (c) $10 \leq x < 40$ (d) $10 < x \leq 40$
- 74.** In a $\triangle ABC$, if $\angle B = 2$, $\angle C = 2\angle A$. Then, what is the ratio of AC to AB ? **2014 II**
 (a) $\sqrt{2} : 1$ (b) $\sqrt{3} : 1$ (c) 1 : 1 (d) $1 : \sqrt{2}$
- 75.** Three straight lines are drawn through the three vertices of a $\triangle ABC$, the line through each vertex being parallel to the opposite side. The $\triangle DEF$ is bounded by these parallel lines. **2014 II**
 Consider the following statements in respect of the $\triangle DEF$.
 I. Each side of $\triangle DEF$ is double the side of $\triangle ABC$ to which it is parallel.
 II. Area of $\triangle DEF$ is four times the area of $\triangle ABC$.
 Which of the statement(s) given above is/are correct?
 (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II
- 76.** In a $\triangle ABC$, AD is the median through A and E is the mid-point of AD and BE produced meets AC at F . Then, AF is equal to **2014 II**
 (a) $AC/5$ (b) $AC/4$ (c) $AC/3$ (d) $AC/2$
- 77.** The point O is equidistant from the three sides of a $\triangle ABC$. Consider the following statements: **2015 II**
 I. $\angle OAC + \angle OCB + \angle OBA = 90^\circ$
 II. $\angle BOC = 2\angle BAC$
 III. The perpendiculars drawn from any point on OA to AB and AC are always equal
 Which of the above statements are correct?
 (a) I and II (b) II and III (c) I and III (d) All of these
- 78.** An equilateral $\triangle BOC$ is drawn inside a square $ABCD$. If $\angle AOD = 2\theta$, what is $\tan \theta$ equal to? **2015 II**
 (a) $2 - \sqrt{3}$ (b) $1 + \sqrt{2}$ (c) $4 - \sqrt{3}$ (d) $2 + \sqrt{3}$
- 79.** In a $\triangle PQR$, point X is on PQ and point Y is on PR such that $XP = 1.5$ units, $XQ = 6$ units, $PY = 2$ units and $YR = 8$ units. Which of the following are correct?
 I. $QR = 5XY$ II. QR is parallel to XY .
 III. $\triangle PYX$ is similar to $\triangle PRQ$.

Select the correct answer using the codes given below.

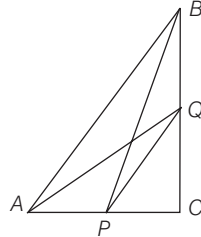
☑ 2016 I

- (a) I and II (b) II and III
(c) I and III (d) All of these

80. ABC is a triangle right angled at C as shown below. Which one of the following is correct?

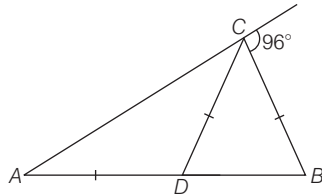
☑ 2016 I

- (a) $AQ^2 + AB^2 = BP^2 + PQ^2$
(b) $AQ^2 + PQ^2 = AB^2 + BP^2$
(c) $AQ^2 + BP^2 = AB^2 + PQ^2$
(d) $AQ^2 + AP^2 = BK^2 + KQ^2$



81. In the given figure, $AD = CD = BC$. What is the value of $\angle CDB$?

☑ 2016 I



- (a) 32° (b) 64°
(c) 78°
(d) Cannot be determined due to insufficient data

82. ABC is an equilateral triangle and X, Y and Z are the points on BC, CA and AB respectively, such that $BX = CY = AZ$. Which of the following is/are correct?

☑ 2016 I

- I. XYZ is an equilateral triangle.
II. $\triangle XYZ$ is similar to $\triangle ABC$.

Select the correct answer using the codes given below.

- (a) Only I (b) Only II (c) Both I and II (d) Neither I nor II

83. Let ABC and $A'B'C'$ be two triangles in which $AB > A'B', BC > B'C'$ and $CA > C'A'$. Let D, E and F be the mid-points of the sides BC, CA and AB , respectively. Let D', E' and F' be the mid-points of the sides $B'C', C'A'$ and $A'B'$, respectively.

Consider the following statements ☑ 2016 I

- I. $AD > A'D', BE > B'E'$ and $CF > C'F'$ are always true.

$$\text{II. } \frac{AB^2 + BC^2 + CA^2}{AD^2 + BE^2 + CF^2} = \frac{A'B'^2 + B'C'^2 + C'A'^2}{A'D'^2 + B'E'^2 + C'F'^2}$$

Which one of the following is correct in respect of the above statements?

- (a) Both statement I and statement II are true and statement II is the correct explanation of statement I
(b) Both statement I and statement II are true but statement II is not the correct explanation of statement I
(c) Statement I is true but statement II is false
(d) Statement I is false but statement II is true

84. ABC and DEF are similar triangles. If the ratio of side AB to side DE is $(\sqrt{2} + 1) : \sqrt{3}$, then the ratio of area of $\triangle ABC$ to that of $\triangle DEF$ is ☑ 2016 I

- (a) $(3 - 2\sqrt{2}) : 3$ (b) $(9 - 6\sqrt{2}) : 2$
(c) $1 : (9 - 6\sqrt{2})$ (d) $(3 + 2\sqrt{2}) : 2$

85. Let $\triangle ABC$ and $\triangle DEF$ be such that $\angle ABC = \angle DEF$, $\angle ACB = \angle DFE$ and $\angle BAC = \angle EDF$. Let L be the mid-point of BC and M be the mid-point of EF .

Consider the following statements ☑ 2016 I

- I. $\triangle ABL$ and $\triangle DEM$ are similar.
II. $\triangle ALC$ is congruent to $\triangle DMF$ even, if $AC \neq DF$.

Which one of the following is correct in respect of the above statements?

- (a) Both statement I and statement II are true and statement II is the correct explanation of statement I
(b) Both statement I and statement II are true but statement II is not the correct explanation of statement I
(c) Statement I is true but statement II is false
(d) Statement I is false but statement II is true

86. ABC is a triangle in which D is the mid-point of BC and E is the mid-point of AD .

- I. The area of $\triangle ABC$ is equal to four times the area of $\triangle BED$.

- II. The area of $\triangle ADC$ is twice the area of $\triangle BED$.

Select the correct answer using the codes given below.

☑ 2016 I

- (a) Only I (b) Only II (c) Both I and II (d) Neither I nor II

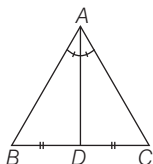
ANSWERS

1	b	2	a	3	b	4	c	5	c	6	d	7	a	8	b	9	c	10	a
11	d	12	b	13	c	14	d	15	d	16	a	17	c	18	a	19	b	20	a
21	d	22	c	23	d	24	a	25	b	26	d	27	a	28	a	29	c	30	d
31	b	32	a	33	d	34	d	35	d	36	c	37	d	38	b	39	c	40	a
41	a	42	d	43	b	44	c	45	d	46	b	47	b	48	c	49	c	50	d
51	b	52	c	53	d	54	a	55	c	56	d	57	d	58	d	59	d	60	d
61	d	62	a	63	d	64	b	65	d	66	b	67	b	68	d	69	b	70	b
71	a	72	a	73	a	74	a	75	c	76	c	77	c	78	d	79	d	80	c
81	b	82	c	83	b	84	c	85	c	86	c								

HINTS AND SOLUTIONS

1. (b) Let ABC is any triangle and AD is the bisector of angle A .

Also, $BD = DC$ (given)



$\therefore AD$ is the internal bisector of $\angle A$.

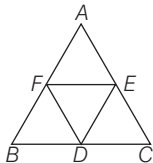
$$\therefore \frac{AB}{AC} = \frac{BD}{DC}$$

$$\Rightarrow \frac{AB}{AC} = \frac{BD}{DC} = 1 \quad [\because BD = DC \text{ given}]$$

$$\Rightarrow AB = AC$$

Hence, triangle is an isosceles triangle.

2. (a) The line segments joining the mid-points of the sides of a triangle form four triangles each of which is similar to the original triangle.



Here, $\triangle BDF \sim \triangle ABC$

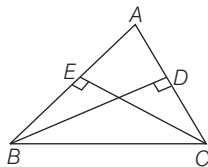
Also, $\triangle DEC, \triangle DEF$

$\triangle AFE \sim \triangle ABC$

3. (b) As $BD = EC$,

$$\angle AEC = \angle BDA = 90^\circ \text{ each}$$

$[\because BD \perp AC \text{ and } CE \perp AB]$



Also, $\angle A = \angle A$ [common]

$\therefore \triangle BDA \cong \triangle AEC$ [by AAS congruency]

$$\Rightarrow AB = AC \quad [\text{by cpct}]$$

So, triangle is an isosceles triangle.

4. (c) Let the other side be b and p .

$$\therefore \frac{1}{2} b \times p = 6 \Rightarrow b \times p = 12$$

$$\Rightarrow b = \frac{12}{p}$$

Also, by pythagoras theorem

$$b^2 = b^2 + p^2$$

$$5^2 = \left(\frac{12}{p}\right)^2 + p^2$$

$$\Rightarrow 25 = \frac{144}{p^2} + p^2$$

$$25p^2 = 144 + p^4$$

$$\Rightarrow p^4 - 25p^2 + 144 = 0$$

$$\Rightarrow p^4 - 16p^2 - 9p^2 + 144 = 0$$

$$\Rightarrow p^2(p^2 - 16) - 9(p^2 - 16) = 0$$

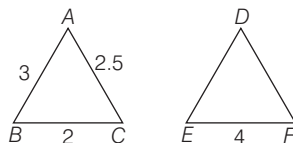
$$\Rightarrow (p^2 - 9)(p^2 - 16) = 0$$

$$\Rightarrow p = 3 \text{ or } p = 4$$

Hence, other sides are 3 cm and 4 cm.

5. (c) As $\triangle ABC \sim \triangle DEF$

$$\Rightarrow \frac{AB}{DE} = \frac{AC}{DF} = \frac{BC}{EF}$$



$$\therefore \frac{BC}{EF} = \frac{2}{4} = \frac{1}{2}$$

$$\therefore DE = 2AB = 2 \times 3 = 6 \text{ cm}$$

$$\text{and } DF = 2 \times AC = 2 \times 2.5 = 5 \text{ cm}$$

$$\therefore \text{Perimeter of } \triangle DEF = (6 + 5 + 4) = 15 \text{ cm}$$

Shortcut Method

Perimeter of $\triangle ABC$

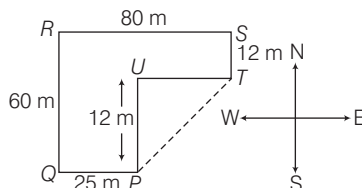
Perimeter of $\triangle DEF$

= Ratio of corresponding sides

$$\therefore \frac{(3 + 2 + 2.5)}{\text{Perimeter of } \triangle DEF} = \frac{1}{2}$$

$$\therefore \text{Perimeter of } \triangle DEF = 2(7.5) = 15 \text{ cm.}$$

6. (d) Let P be the starting point of his run, then PT is the distance between the starting and the finishing point.



$$\therefore PU = RQ - ST = 60 - 12 = 48 \text{ m}$$

$$\text{and } UT = RS - QP = 80 - 25 = 55 \text{ m}$$

$$\therefore \text{In } \triangle PUT, PT^2 = (PU)^2 + (TU)^2$$

$$\therefore PT = \sqrt{(48)^2 + (55)^2} = \sqrt{2304 + 3025}$$

$$= \sqrt{5329} = 73 \text{ m}$$

7. (a) As ADB is a right angled triangle.

$$\text{So, } AB^2 = AD^2 + BD^2$$

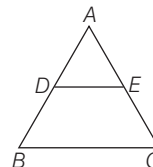
$$\Rightarrow AC^2 = AD^2 + BD^2 \quad [\because AB = AC]$$

$$\Rightarrow (AD + DC)^2 = AD^2 + BD^2$$

$$\Rightarrow AD^2 + DC^2 + 2AD \cdot DC = AD^2 + BD^2$$

$$\Rightarrow BD^2 - CD^2 = 2DC \cdot AD$$

8. (b) As in $\triangle ADE$ and $\triangle ABC$,
 $\frac{AD}{AB} = \frac{8}{20} = \frac{2}{5}$ and $\frac{AE}{AC} = \frac{6}{15} = \frac{2}{5}$



$$\therefore \frac{AD}{AB} = \frac{AE}{AC}$$

and $\angle A = \angle A$ [common]

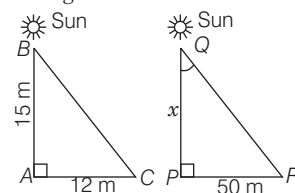
$\therefore \triangle ADE \sim \triangle ABC$

[by SAS similarity]

$$\therefore \frac{DE}{BC} = \frac{AD}{AB} \Rightarrow \frac{DE}{BC} = \frac{2}{5}$$

$$\Rightarrow BC = \frac{5}{2} DE.$$

9. (c) Let AB be a vertical stick and AC be its shadow. Also, let PQ be a tower having shadow PR .



$$\angle A = \angle P$$

$$\angle B = \angle Q$$

$[\because \text{Sun with tower and stick forms same angle}]$

As, $\triangle ABC \sim \triangle PQR$

[$\therefore$ By AA similarity]

$$\therefore \frac{AB}{PQ} = \frac{AC}{PR} \Rightarrow \frac{15}{x} = \frac{12}{50}$$

$$\Rightarrow x = \frac{15 \times 50}{12} = 62.5 \text{ m}$$

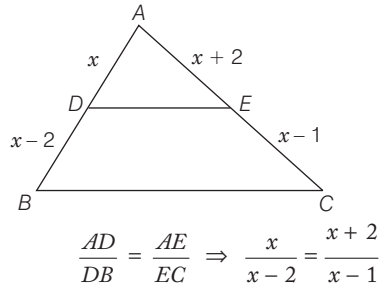
Hence, the height of the tower is 62.5 m.

10. (a) Let the ratio of their corresponding height be $h_1 : h_2$.

But the ratio of the areas of two similar triangles is equal to the ratio of the squares of their corresponding heights.

$$\therefore \frac{h_1^2}{h_2^2} = \frac{81}{49} \Rightarrow h_1 : h_2 = 9 : 7$$

11. (d) As $DE \parallel BC$, by basic proportionality theorem,



$$\begin{aligned} \Rightarrow x(x-1) &= (x+2)(x-2) \\ \Rightarrow x^2 - x &= x^2 - 2x + 2x - 4 \\ \Rightarrow x &= 4 \end{aligned}$$

Hence, the value of x is 4.

12. (b) As $AB \parallel CD$ and the diagonals of a trapezium divide each other proportionally.

$$\begin{aligned} \text{So, } \frac{AO}{OC} &= \frac{BO}{OD} \\ \Rightarrow \frac{3x-1}{5x-3} &= \frac{2x+1}{6x-5} \end{aligned}$$

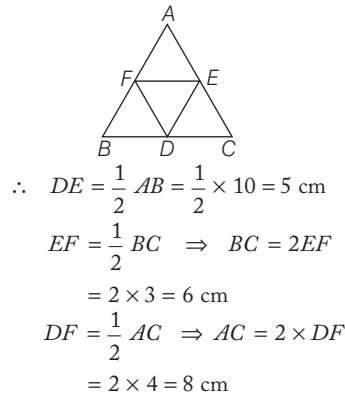
$$\begin{aligned} (3x-1)(6x-5) &= (5x-3)(2x+1) \\ \Rightarrow 18x^2 - 15x - 6x + 5 &= 10x^2 + 5x - 6x - 3 \\ \Rightarrow 8x^2 - 20x + 8 &= 0 \\ \Rightarrow 4x^2 - 10x + 4 &= 0 \\ \Rightarrow 2(2x^2 - 5x + 2) &= 0 \\ \Rightarrow 2(2x^2 - 4x - x + 2) &= 0 \\ \Rightarrow 2[2x(x-2) - 1(x-2)] &= 0 \\ \Rightarrow 2(2x-1)(x-2) &= 0 \\ \text{But as } x = \frac{1}{2} \text{ will make } OC \text{ negative.} \\ \therefore x &= 2 \end{aligned}$$

13. (c) Exterior angle bisector theorem The exterior bisector of an angle of a triangle divides the opposite side externally in the ratio of the sides containing the angle.

$$\therefore \frac{BE}{CE} = \frac{AB}{AC} \text{ as } AE \text{ is an exterior angle bisector.}$$

$$\begin{aligned} \text{Let } CE &= x, BE = BC + EC = 12 + x \\ \Rightarrow \frac{12+x}{x} &= \frac{10}{6} \\ \Rightarrow (12+x)6 &= 10x \\ \Rightarrow 72 + 6x &= 10x \Rightarrow 4x = 72 \\ \Rightarrow x &= 18 \text{ cm} \end{aligned}$$

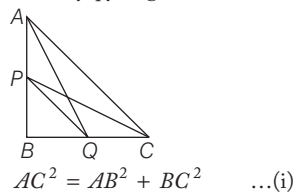
14. (d) As the line joining the mid-points of any two sides of a triangle is parallel to the third side and is half of the third side.



15. (d) In $\triangle PQR$, $\angle P + \angle Q + \angle R = 180^\circ$
 $\Rightarrow a + 3a + b = 180^\circ$
 $\Rightarrow 4a + b = 180^\circ \dots(i)$
 Given, $-5a + 3b = 30^\circ \dots(ii)$
 Solving Eqs. (i) and (ii), we get $a = 30^\circ$ and $b = 60^\circ$
 $\therefore \angle P = 30^\circ, \angle Q = 90^\circ$ and $\angle R = 60^\circ$
 So, $\triangle PQR$ is right angled triangle.

16. (a) Consider $\triangle BMD$ and $\triangle DLC$,
 As $\angle BMD = \angle DLC = 90^\circ$ [each]
 Also, $\angle BDM = \angle DCL$ corresponding angle
 $\therefore \triangle BMD \sim \triangle DLC$ [by AA similarity]
 $\therefore \frac{BD}{DC} = \frac{DM}{LC} = \frac{BM}{DL}$
 $\Rightarrow \frac{BD}{DC} = \frac{DM}{LC} = \frac{1}{3}$
 $\therefore DM : LC = 1 : 3$

17. (c) In $\triangle ABC$, by pythagoras theorem,



$$\begin{aligned} \text{And in } \triangle PBQ, \\ PQ^2 &= PB^2 + BQ^2 \dots(ii) \\ \text{Adding Eqs. (i) and (ii),} \\ AC^2 + PQ^2 &= (AB^2 + BC^2) + PB^2 + BQ^2 \\ &= (AB^2 + BQ^2) + (PB^2 + BC^2) \\ AC^2 + PQ^2 &= AQ^2 + CP^2 \end{aligned}$$

18. (a) In $\triangle BCA$, $\angle CAB = 65^\circ$ and $\angle ACB = 90^\circ$ [$\because \angle C = 90^\circ$ given]
 $\angle CBA = 180^\circ - (\angle BCA + \angle CAB)$
 [by angle sum property of a triangle]
 $= 180^\circ - (65^\circ + 90^\circ)$
 $= 180^\circ - 155^\circ = 25^\circ$

19. (b) Let the angles of a triangle be $2x, 3x, 4x$, then
 $2x + 3x + 4x = 180^\circ$
 [by angle sum property of a triangle]
 $9x = 180^\circ \Rightarrow x = 20^\circ$

$$\begin{aligned} \text{So, angles are } 2x &= 40^\circ, \\ 3x &= 60^\circ, 4x = 80^\circ. \end{aligned}$$

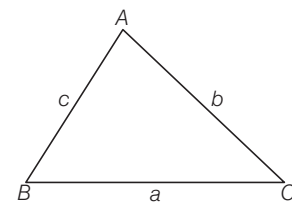
20. (a) In $\triangle ABC$, $\angle A + \angle B + y = 180^\circ$
 $y = 180^\circ - (40 + 30)^\circ = 110^\circ$
 $\angle ADE = \angle ABC$ [corresponding angle]
 $x = 30^\circ$
 Similarly, $y = z = 110^\circ$

21. (d) In $\triangle GEF$ and $\triangle GCD$, we have
 $\angle EFG = \angle GDC$ [alternate angles]
 $\angle EGF = \angle CGD$
 [vertically opposite angles]
 $\triangle GEF \sim \triangle GCD$ [by AA similarity]

$$\begin{aligned} \therefore \frac{GE}{CG} &= \frac{EF}{DC} \\ \Rightarrow \frac{5}{10} &= \frac{EF}{18} \\ \Rightarrow EF &= \frac{5 \times 18}{10} = 9 \text{ cm} \end{aligned}$$

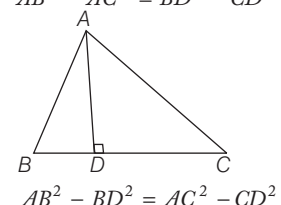
22. (c) Given, In $\triangle PQR$, $PQ > PR$
 $\Rightarrow \angle PRQ > \angle PQR$
 [angle opp to larger side is greater]
 $\Rightarrow \frac{1}{2} \angle PRQ > \frac{1}{2} \angle PQR$
 $\Rightarrow \angle SRQ > \angle SQR \Rightarrow SQ > SR$
 [side opp to greater angle is larger]

23. (d) Given, $AB = AC$

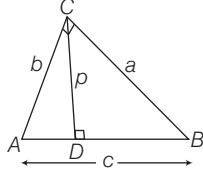


$$\begin{aligned} \Rightarrow \angle ABC &= \angle ACB \\ \text{If } \angle A &= 60^\circ, \text{ then } \triangle ABC \text{ is an equilateral triangle } a = b = c \\ \text{If } \angle A &= 90^\circ, \text{ then } \triangle ABC \text{ is right angled triangle} \\ a^2 &= b^2 + c^2 = c^2 + c^2 = 2c^2 [\because b = c] \\ \Rightarrow a &= \sqrt{2} c \\ \therefore \text{ If } 60^\circ < A < 90^\circ, \text{ then } c < a < c\sqrt{2} \end{aligned}$$

24. (a) Here, in $\triangle ABC$
 $AB^2 = AD^2 + BD^2 \dots(i)$
 In right angled $\triangle ADC$, we have
 $AC^2 = AD^2 + CD^2 \dots(ii)$
 Subtracting Eq. (ii) from Eq. (i),
 $AB^2 - AC^2 = BD^2 - CD^2$



25. (b) Since, c is the base and p is the altitude of $\triangle ABC$.

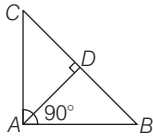


Here, area of $\triangle ABC = \frac{1}{2} pc$... (i)

Also, area of $\triangle ABC = \frac{1}{2} ab$... (ii)

From Eqs. (i) and (ii), we get
 $\frac{1}{2} pc = \frac{1}{2} ab \Rightarrow pc = ab$

26. (d) Hence, $AD^2 = BD \cdot DC$



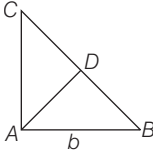
$$\Rightarrow \frac{AD}{BD} = \frac{DC}{AD}$$

$\therefore \triangle ADB \sim \triangle CDA \Rightarrow \angle BAD = \angle ACD$

Now, $\angle CAB = \angle CAD + \angle BAD$
 $= \angle CAD + \angle ACD = 90^\circ$

So, $\triangle ABC$ must be right angled triangle.
Hence, $BC^2 = AC^2 + AB^2$

27. (a) In $\triangle ABC$,



Area of $\triangle ABC = \frac{1}{2} \times \text{Base} \times \text{Altitude}$

$$\Delta = \frac{1}{2} b \times AC, AC = \frac{2\Delta}{b}$$

In $\triangle ABC$, Using pythagoras theorem,
 $AC^2 + AB^2 = BC^2$

$$\Rightarrow BC = \sqrt{\frac{4\Delta^2}{b^2} + b^2}$$

Again in $\triangle ABC$, area of $\triangle ABC$

$$\Delta = \frac{1}{2} \times BC \times AD$$

$$\Rightarrow AD = \frac{2\Delta}{BC} = \frac{2\Delta b}{\sqrt{4\Delta^2 + b^4}}$$

28. (a) As, $AB \parallel CD$ and BC cuts them.
 $\therefore \angle BCD = \angle ABC = x$ [alternate $\angle$ s]

$$\therefore \text{In } \triangle BCD, x + y + z = 180^\circ$$

$$\frac{4}{3}y + y + z = 180^\circ$$

$$\Rightarrow \frac{7y}{3} + z = 180^\circ$$

$$\frac{7}{3} \left(\frac{3}{8}z \right) + z = 180^\circ \Rightarrow \frac{7}{8}z + z = 180^\circ$$

$$\left[\because y = \frac{3}{8}z \right]$$

$$\frac{15z}{8} = 180^\circ \Rightarrow z = 96^\circ$$

$$\text{So, } y = \frac{3}{8} \times 96^\circ = 36^\circ$$

$$\text{and } x = \frac{4}{3} \times 36^\circ = 48^\circ$$

29. (c) We have, $x = 35^\circ$ [alternate angles]
 $z = 75^\circ$ [alternate angles]

In $\triangle ABP$, $x + y + 75^\circ = 180^\circ$

$$y = 180^\circ - (75^\circ + 35^\circ) \Rightarrow y = 70^\circ$$

Hence the value of

$$x = 35^\circ, y = 70^\circ, z = 75^\circ$$

30. (d) $x^\circ = 60^\circ$ [vertically opposite angles]
 $y = 60^\circ$ [corresponding angles]

$\angle PRS = 110^\circ$ [alternate angles]

$\angle QRS + x^\circ = 110^\circ$ [Alternate angles]

$$\angle QRS = 110^\circ - 60^\circ = 50^\circ$$

$$\therefore t = 180^\circ - (y + \angle QRS)$$

$$= 180^\circ - (60^\circ + 50^\circ)$$

$$t = 70^\circ$$

Also, $t = z$ [alternate angles]

$$\therefore z = 70^\circ$$

31. (b) Since, $\angle PTB = 55^\circ$ [given]
Then, $\angle TUV = 55^\circ$

[corresponding angle]

Also, $\angle TUV = \angle CUQ = 55^\circ$

[vertically opposite angle]

Also given, $\angle DVS = 45^\circ$

Then, $\angle UVT = 45^\circ$

[vertically opposite angle]

In $\triangle UTV$,

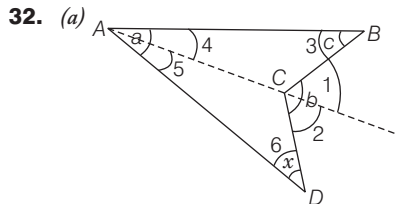
$$\angle T = 180^\circ - (55^\circ + 45^\circ) = 80^\circ$$

[angle sum property of a triangle]

$$\Rightarrow \angle T = \angle PTR = 80^\circ$$

[vertically opposite angle]

$$\therefore \angle CUQ + \angle RTP = 55^\circ + 80^\circ = 135^\circ$$



$$\angle 1 = \angle 3 + \angle 4, \angle 2 = \angle 5 + \angle 6$$

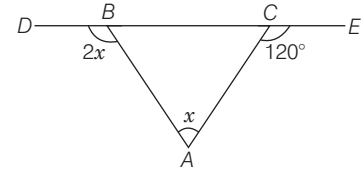
$$\Rightarrow \angle 1 + \angle 2 = \angle 3 + \angle 4 + \angle 5 + \angle 6$$

[since, exterior angle is equal to sum of two opposite interior angles]

$$\Rightarrow b = c + a + x$$

$$\therefore x = b - c - a$$

33. (d) $\therefore \angle ABC = 180^\circ - \angle DBA = 180^\circ - 2x$



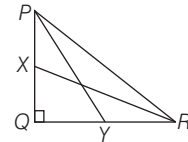
and $\angle ACB = 180^\circ - \angle ACE$

$$= 180^\circ - 120^\circ = 60^\circ$$

In $\triangle ABC$, $\angle ABC + \angle ACB + \angle BAC = 180^\circ$

$$\Rightarrow 180^\circ - 2x + 60^\circ + x = 180^\circ \Rightarrow x = 60^\circ$$

34. (d) In $\triangle PQY$, by pythagoras theorem,
 $PY^2 = PQ^2 + QY^2$



$$\Rightarrow PY^2 = PQ^2 + \left(\frac{QR}{2} \right)^2$$

$$\left[QY = YR = \frac{QR}{2} \right] \dots (i)$$

and in $\triangle XQR$, $RX^2 = QX^2 + QR^2$

$$\Rightarrow RX^2 = \left(\frac{PQ}{2} \right)^2 + QR^2$$

$$\left[\because PX = XQ = \frac{PQ}{2} \right] \dots (ii)$$

On adding Eqs. (i) and (ii), we get

$$PY^2 + RX^2 = \frac{5PQ^2}{4} + \frac{5QR^2}{4}$$

$$\Rightarrow PY^2 + RX^2 = 5/4(PQ^2 + QR^2)$$

$$\Rightarrow 4(PY^2 + RX^2) = 5(PR^2)$$

In $\triangle XQR$, by pythagoras theorem,

$$XR^2 = XQ^2 + QR^2 \dots (iii)$$

and In $\triangle PQY$, by pythagoras theorem,

$$PY^2 = PQ^2 + QY^2 \dots (iv)$$

On adding Eqs. (iii) and (iv),

$$XR^2 + PY^2 = XQ^2 + QR^2$$

$$+ PQ^2 + QY^2$$

$$\Rightarrow XR^2 + PY^2 = XQ^2 + QY^2$$

$$+ QR^2 + PQ^2$$

$$\Rightarrow XR^2 + PY^2 = XY^2 + PR^2$$

and again,

$$XR^2 + PY^2 = XY^2 + PR^2$$

$$= XY^2 + (2XY)^2$$

$$XR^2 + PY^2 = XY^2 + 4XY^2$$

$$\Rightarrow XR^2 + PY^2 = 5XY^2$$

Thus, option (d) is incorrect.